Dror Kosowsky-Sachs

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Education

University of Maryland, College Park — Expected Graduation May 2027
Major: **Mechanical Engineering**, GPA: **3.89** — College Park, MD
Minor: **Robotics and Autonomous Systems**
Academic Achievements: ASPIRE Scholarship (Summer 2026), Dean's List (Every Semester)

Skills

CAD/CAM & Design: SolidWorks, Autodesk Fusion (CAM), KiCad (PCB design)
Fabrication & Prototyping: CNC Milling, Waterjet, Laser Cutting, 3D Printing, Soldering, Silicone Molding
Programming & Robotics: Python, MATLAB, ROS2 (Nav2, SLAM), Git, Linux, Raspberry Pi, Arduino

Work Experience

Undergraduate Researcher, Hydrodynamics Lab, University of Maryland | June 2025-Present

Biomimetic UUV with 4-DOF Tail Manipulator | May 2026 - Present

- Developed a real-time inverse kinematics solver, reducing per-step computation time from 700 ms to 6 ms to enable smooth servo control
- Designed a 4-DOF manipulator to actuate a robotic fish's tail for greater efficiency and maneuverability, iterating from PLA prototypes to machined aluminum
- Machined 20 structural components to sub-mm tolerances using CNC mill, waterjet, and bandsaw
- Designed and fabricated a custom PCB power converter in KiCad supplying 5 V compute and 7.4 V servo rails
- Wrote ROS2 nodes for fin gait generation, inverse kinematics, and servo control
- Waterproofed the vehicle with multi-layer sealing using 3M sealant, silicone, and gaskets

Flexible Tail Stiffness Project | June 2025 - August 2025

- Investigated flipper-based propulsion for autonomous underwater vehicles over a 10-week research program
- Iterated the fish's CAD model and manufactured all of its components
- Redesigned the silicone casting mold and cast the flexible tail and flipper
- Designed and ran experiments comparing 6 waterproofing methods to identify the most effective solution
- Scaled the full vehicle down to ~30% of its original size to improve swimming efficiency and shorten design iteration time

Internship at Six Degrees of Space | June 2024 - July 2024

- Modeled a high-speed, close-range drone location sensor to support the company's digital twinning efforts
- Designed and executed parametric tests across distance, orientation, and filtering conditions to gain sensor data
- Developed a predictive model to quantify sensor–beacon signal strength as a function of four physical parameters
- Evaluated Sphera digital twinning software and recommended an approach for simulating the company's sensor

Other Projects

Autonomous Chess Board — October 2024-Present

- Designed and fabricated a mechatronic chess board that plays against a human opponent, using a CoreXY gantry with magnetic actuation to move pieces in any direction
- Integrated Raspberry Pi control of servo motors and gantry motion with a computer vision system for move detection
- Built a custom chess clock with limit switch inputs and alternating display to signal turn completion to the computer

Autonomous Maze Navigation — **Spring 2026**

- Programmed TurtleBot 4 robots in ROS2 to escape mazes using LiDAR, in mapped and unmapped races
- Mapped a maze with SLAM in a 5-minute window, then localized with AMCL and used Nav2 to autonomously navigate to goal poses
- Developed a right-hand wall-following state machine with geometric heading estimation and collision-avoidance cases for unmapped navigation
- Validated Nav2 in a 2D TurtleBot simulation before hardware testing, and analyzed performance from recorded rosbags